

Portfolio Question - Behind Boosting Trees

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Portfolio 03.2.P (v4.0.0)

Portfolio Question 3.2.P - Behind Boosting Trees

- ▶ How are the state-of-the-art boosting algorithms implemented?
- ▶ What is the intuition behind why boosting works?

LightGBM and Histogram Boosting

- ▶ LightGBM¹ is a popular, efficient implementation of boosted decision trees
 - ▶ It uses a histogram approach to speed up learning
 - ▶ It is very fast, and can be faster than random forests
- ▶ Speed and efficiency comes from:
 - ▶ Binning continuous features into discrete bins
 - ▶ Binning categorical features into **binary** bins (no one-hot encoding)
 - ▶ Using a leaf-wise tree growth algorithm (over depth-wise)

¹LightGBM: A Highly Efficient Gradient Boosting Decision Tree (2017)
Microsoft Research, [url](#).

Intuition

- ▶ From [Medium: The Intuition Behind Gradient Boosting & XGBoost](#):
 - ▶ If we can account for our model's errors, we will be able to improve our model's performance.
 - ▶ We can train a new predictor to predict the errors made by the original model.
 - ▶ Repeat for each datapoint. Careful to regularise.

Boosting loss function

- ▶ Use the squared error $0.5(y - \hat{y})^2$ with gradient $\hat{y} - y$
- ▶ We're learning some parameter(s) θ
- ▶ When the loss is $\min_{\theta} \sum_{i=1}^N 0.5(y_i - \hat{y}_i(\theta, \mathbf{x}_i))^2 + 0.5\lambda\theta^2$
- ▶ We can solve for θ by taking derivatives and setting equal to zero
- ▶ Note it simplifies if $\hat{y}(\theta)$ is simple
 - ▶ they use the mean in the article
 - ▶ also straightforward for the effect of one covariate on y

Discussion

- ▶ What is the role of λ ?
 - ▶ When θ is small? large?
 - ▶ Does it relate to information criteria?
 - ▶ Are there other models that focus on residuals? Why?